```
pip install torch transformers datasets peft accelerate
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/pyt
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/p
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/p
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-pa
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Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12,
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/d
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/py
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/d
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyth
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12,
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Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packa
```

```
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.:
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/pytho
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-pa
```

```
import torch
import pandas as pd
from datasets import Dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassificatior
from transformers import TrainingArguments, Trainer
from peft import LoraConfig, get_peft_model
```

```
model_name = "distilbert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Userw
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access publ
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated req
config.json: 100%                                                   483/483 [00:00<00:00, 5.01kB/s]

tokenizer_config.json: 100%                                         48.0/48.0 [00:00<00:00, 604B/s]

vocab.txt:        232k/? [00:00<00:00, 1.77MB/s]

tokenizer.json:        466k/? [00:00<00:00, 3.62MB/s]

model.safetensors: 100%                                         268M/268M [00:02<00:00, 185MB/s]

Loading weights: 100%  100/100 [00:00<00:00, 233.90it/s, Materializing param=distilbert.transformer.

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; r
- MISSING   :those params were newly initialized because missing from the checkp
```

```
data = pd.read_csv("sentiment_dataset_peft_lab19.csv")

dataset = Dataset.from_pandas(data)

def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

tokenized_dataset = dataset.map(tokenize_function)

dataset = tokenized_dataset.train_test_split(test_size=0.2)

train_dataset = dataset["train"]
test_dataset = dataset["test"]
```

```
Map: 100%                                    300/300 [00:01<00:00, 507.84 examples/s]
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)

peft_model.print_trainable_parameters()
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Tri
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
)

trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: User
  super().__init__(loader)
```

[26/30 07:52 < 01:18, 0.05 it/s, Epoch 0.83/1]

Step	Training Loss
10	0.688281
20	0.575099

```
trainer.evaluate(eval_dataset=test_dataset)

text = "The movie was fantastic and inspiring!"

inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)

prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```